Spectral Modeling and Power Analysis of Underwater Wireless Optical Communication

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Introduction

Underwater Wireless Optical Communication (UWOC) performance depends on:

- Wavelength-dependent absorption
- Wavelength-dependent scattering
- Environmental parameters
- Transmitter optical efficiency

Goal: Identify optimal wavelength in blue–green region.

Spectral Channel Modeling

Total attenuation (extinction) coefficient:

$$c(\lambda) = a(\lambda) + b(\lambda)$$

Where:

$$a(\lambda) = a_w(\lambda) + a_{ph}(\lambda) + a_g(\lambda) + a_d(\lambda)$$

$$b(\lambda) = b_w(\lambda) + b_p(\lambda)$$

- $a(\lambda)$: Total absorption coefficient (m^{-1})
- $b(\lambda)$: Total scattering coefficient (m^{-1})
- $c(\lambda)$: Beam attenuation coefficient (m^{-1})

Absorption Coefficient Model

$$a(\lambda) = a_w(\lambda) + a_{ph}(\lambda) + a_g(\lambda) + a_d(\lambda)$$

- $a_w(\lambda)$: Pure water absorption coefficient
- $a_{ph}(\lambda)$: Phytoplankton absorption (chlorophyll dependent)
- $a_g(\lambda)$: Gelbstoff (CDOM / humic dissolved organic matter)
- $a_d(\lambda)$: Detritus absorption

Chlorophyll-dependent absorption:

$$a_{ph}(\lambda) = A_{ph}(\lambda) \cdot C_{chl}$$

- C_{chl} : Chlorophyll concentration (mg/m^3)

Scattering Coefficient Model

$$b(\lambda) = b_w(\lambda) + b_p(\lambda)$$

- $b_w(\lambda)$: Molecular (Rayleigh) scattering by pure water
- $b_p(\lambda)$: Particulate scattering coefficient

Pure water scattering:

$$b_w(\lambda) \propto \lambda^{-4.32}$$

Particle scattering model:

$$b_p(\lambda) = B_p(\lambda) \cdot C_p$$

- C_p : Particle concentration

Propagation Model

Beer–Lambert law:

$$P_r(\lambda, d) = P_t(\lambda) \exp\left[-c(\lambda)d\right]$$

Where:

- P_t : Transmitted optical power
- P_r : Received optical power
- d : Propagation distance (m)
- $c(\lambda)$: Total attenuation coefficient

Optical Transmitter Model

Electrical input power:

$$P_{elec} = VI$$

Optical transmitted power:

$$P_t = \eta_{eo} \cdot P_{elec}$$

Where:

- η_{eo} : Electrical-to-optical efficiency
- V : Drive voltage
- I : Drive current

Absorption Coefficient vs Wavelength

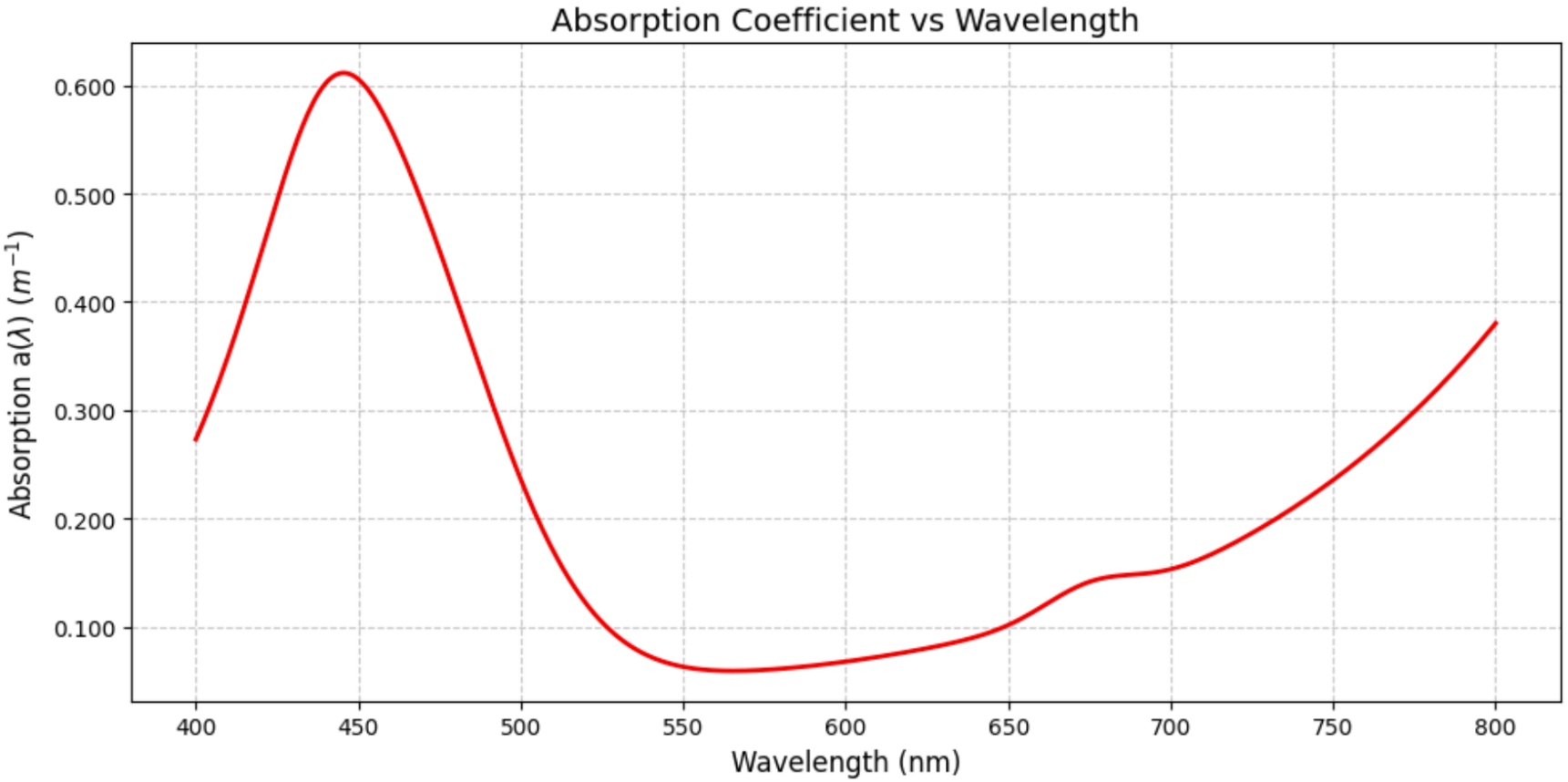

Figure 1: Absorption Coefficient vs Wavelength

Scattering Coefficient vs Wavelength

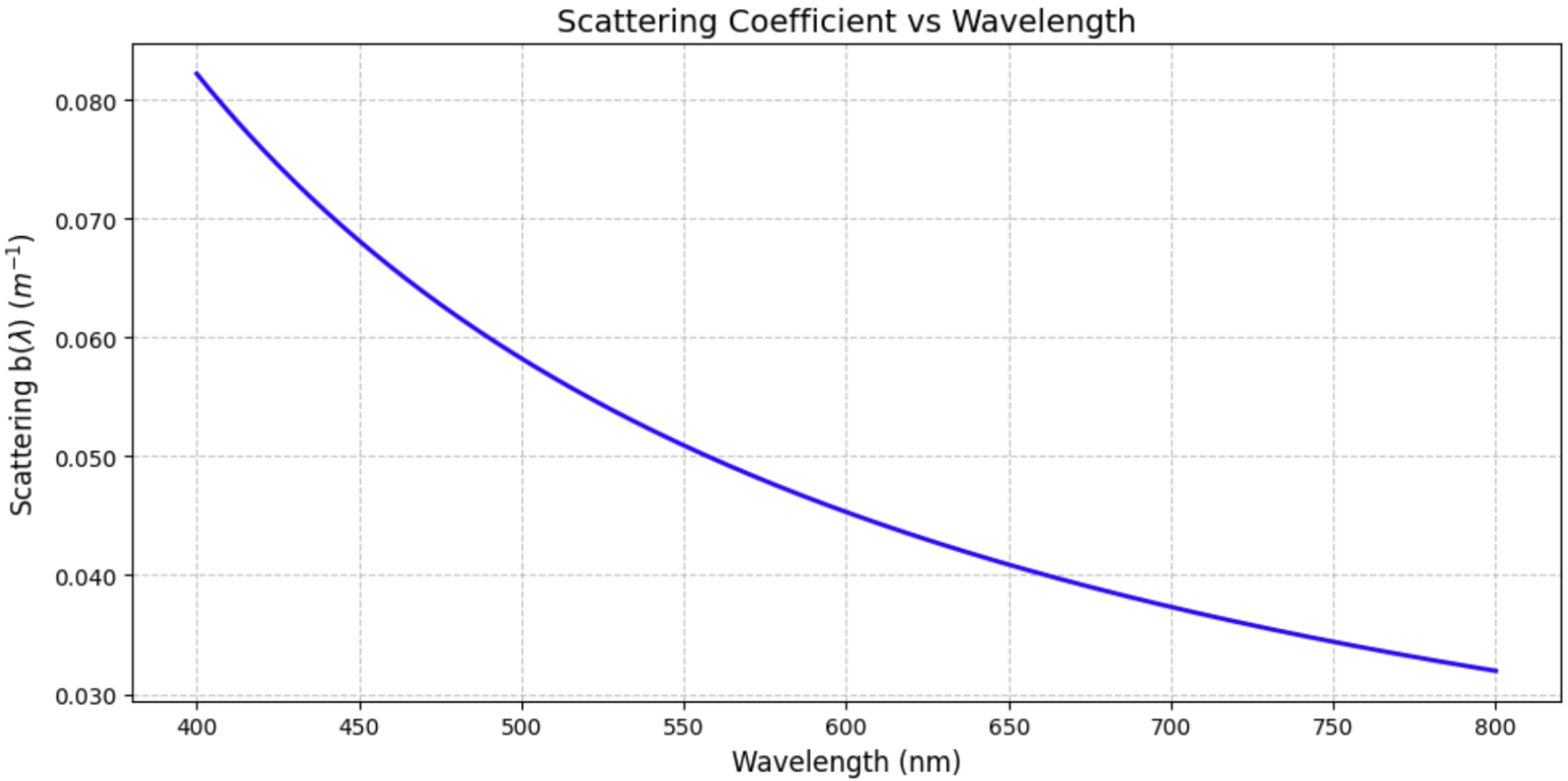

Figure 2: Scattering Coefficient vs Wavelength

Total Attenuation Coefficient vs Wavelength

$$c(\lambda) = a(\lambda) + b(\lambda)$$

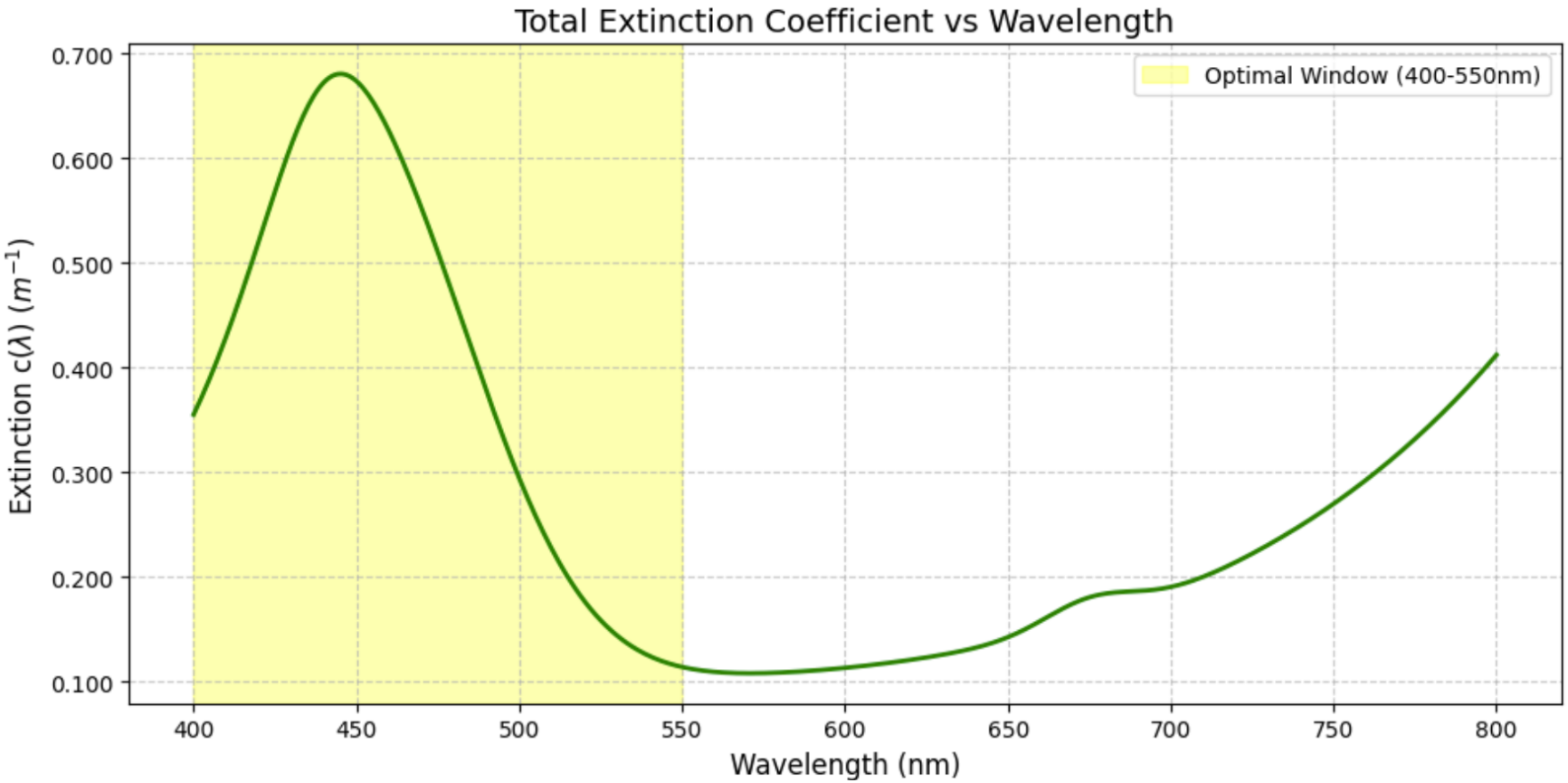

Figure 3: Total Attenuation Coefficient vs Wavelength

Blue vs Green Discrete Comparison

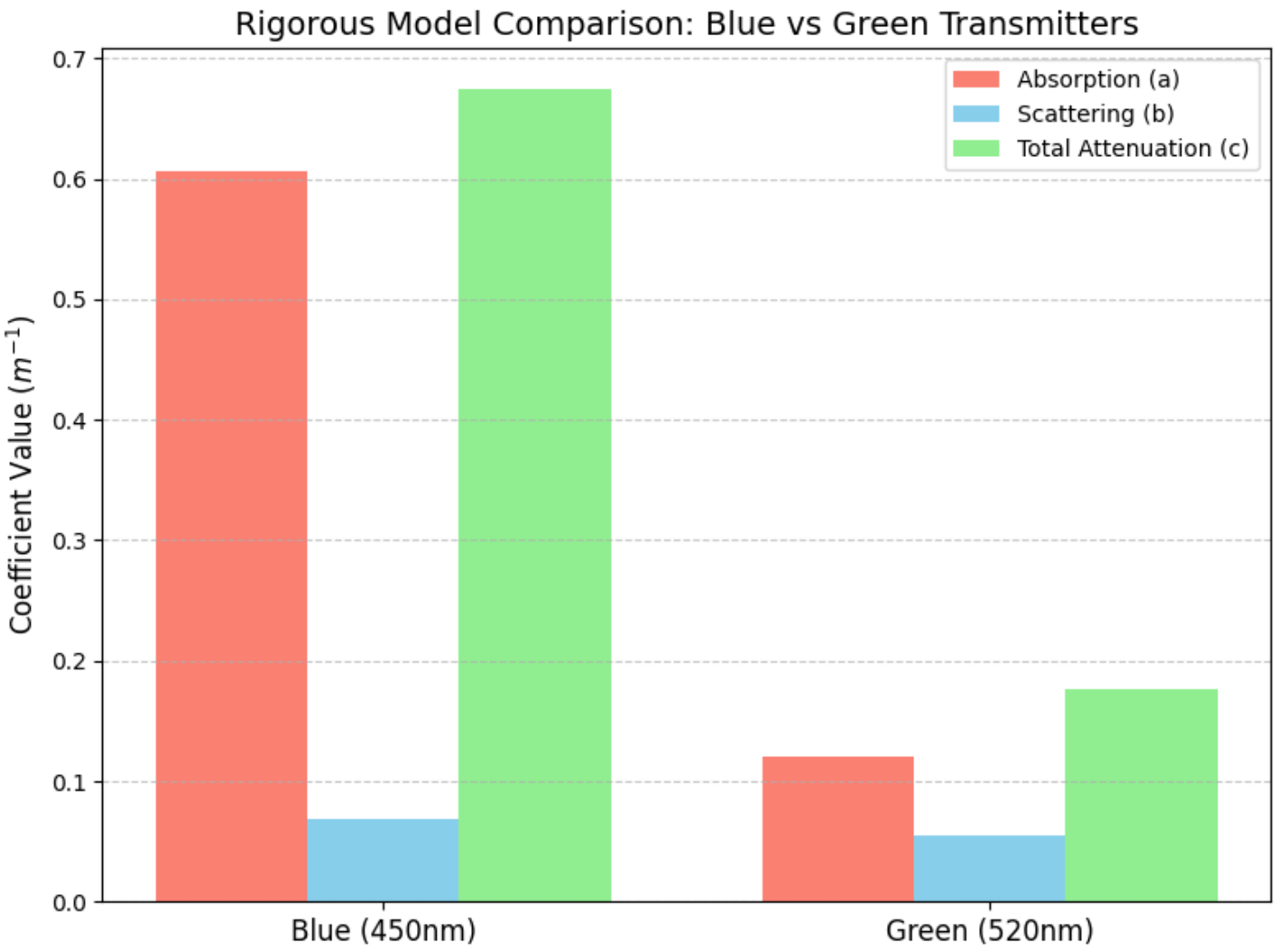

Figure 4: Blue vs Green Performance Comparison

Key Observations

- Minimum attenuation occurs in 400–550 nm window.
- Blue: Lower absorption, higher scattering.
- Green: Slightly higher absorption, lower scattering.
- Environmental concentration strongly impacts performance.

Conclusion

- Continuous spectral modeling gives realistic UWOC behavior.
- Blue–green region is optimal for underwater transmission.
- Transmitter efficiency directly affects optical power.
- Model can be extended using Beer–Lambert law for distance loss.